

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 39-2021: A 26-Year-Old Woman with Respiratory Failure and Altered Mental Status

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PRESENTATION OF CASE

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Dr. Viral S. Shah: A 26-year-old woman was transferred to this hospital because of respiratory failure and altered mental status.

The patient had been in her usual state of health until 3 days before the current admission, when she underwent elective abdominoplasty and liposuction at an ambulatory surgery center. During the procedure, anesthetic agents included intravenous propofol, fentanyl, midazolam, and ketamine, as well as local lidocaine injections. During recovery after the procedure, medications included intravenous cefazolin, enoxaparin, and ondansetron; oral oxycodone, tramadol, and gabapentin; and transdermal scopolamine. The patient was lethargic after the procedure but was able to be discharged home after a period of observation.

Later that day, the patient was instructed by telephone to take a lower dose of postoperative oxycodone. The next day, on evaluation at the surgical center, she was found to be somnolent but arousable and able to answer questions with effort; the scopolamine patch was removed. Several hours after the patient returned home, her family noticed that she had increased respiratory effort and minimal responsiveness. She was taken by ambulance to the emergency department of a local hospital.

In the emergency department, the temperature was 37.7°C, the heart rate 130 beats per minute, the blood pressure 121/77 mm Hg, the respiratory rate 30 breaths per minute, and the oxygen saturation 83% while the patient was breathing ambient air. She was lethargic, moaned intermittently, and responded with one-word answers; symmetric movement in the arms and legs was observed. Intravenous naloxone was administered, with minor improvement in her condition. Levels of electrolytes and results of kidney- and liver-function tests were normal; other laboratory test results are shown in Table 1. An electrocardiogram showed sinus tachycardia, rightward axis deviation, and nonspecific ST-T-wave changes.

Dr. Dexter P. Mendoza: A chest radiograph showed diffuse confluent opacities in both lungs (Fig. 1A). Computed tomography (CT) of the chest, performed after the

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, Other Hospital	1 Day before Transfer, Other Hospital	Reference Range, Adults, This Hospital†	On Arrival, This Hospital
Hemoglobin (g/dl)	12.0–15.2	9.1	12.0–16.0	9.2
Hematocrit (%)	34.1–44.9	26.3	36.0–46.0	27.6
White-cell count (per μ l)	3900–10,800	21,800	4500–11,000	17,670
Differential count (%)				
Neutrophils	43–74	82	40–70	92.7
Lymphocytes	17–44	8.4	22–44	4.4
Monocytes	4–13	8.3	4–11	1.7
Eosinophils	1–10	0.6	0–8	0.0
Platelet count (per μ l)	154,000–369,000	201,000	150,000–400,000	221,000
Albumin (g/dl)	2.5	3.5–4.8	3.3–5.0	3.1
Lactate (mmol/liter)	0.5–2.2	4.5	0.5–2.2	1.6
Procalcitonin (ng/ml)	<0.10	3.09	0.00–0.08	1.59
Troponin I (ng/ml)	<0.029	0.65	—	—
High-sensitivity troponin T (ng/liter)	—	—	<0.03	237
B-type natriuretic peptide (pg/ml)	0–100	395	—	—
N-terminal pro-B-type natriuretic peptide (pg/ml)	—	—	<450	1865
Erythrocyte sedimentation rate (mm/hr)	—	—	0–13	63
C-reactive protein (mg/liter)	—	—	<8	204.8
Triglycerides (mg/dl)	—	—	0–150	167
Arterial blood gas				
Fraction of inspired oxygen	—	—	—	1.00
pH	—	—	7.35–7.45	7.44
Partial pressure of carbon dioxide (mm Hg)	—	—	35–42	39
Partial pressure of oxygen (mm Hg)	—	—	80–100	116

* To convert the values for lactate to milligrams per deciliter, divide by 0.1110. To convert the values for triglycerides to millimoles per liter, multiply by 0.01129.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

administration of intravenous contrast material as part of a pulmonary-embolism protocol, ruled out pulmonary embolism but revealed diffuse mixed ground-glass and consolidative opacities involving all lobes of the lungs, with relative sparing of the subpleural lung (Fig. 1B). CT of the abdomen revealed postoperative changes. CT of the head, performed without the administration of contrast material, revealed mild effacement of the cerebral sulci and cisterns on both sides, a finding suggestive of mild diffuse cerebral edema (Fig. 1C and 1D). There was also a small hypodensity in the left globus pallidus, a finding suggestive of infarction.

Dr. Shah: Noninvasive bilevel positive airway pressure was initiated, and the patient was admitted to the intensive care unit (ICU) at the other hospital. Intravenous vancomycin, piperacillin-tazobactam, levofloxacin, and oseltamivir were administered empirically, and a triple-lumen central venous catheter was inserted.

The next morning, the findings on a repeat chest radiograph appeared to be unchanged, but the patient had progressive lethargy, so intubation was performed and mechanical ventilation was initiated. Pink frothy secretions were noted. Intravenous methylprednisolone, propofol, and norepinephrine were administered. A nucleic acid

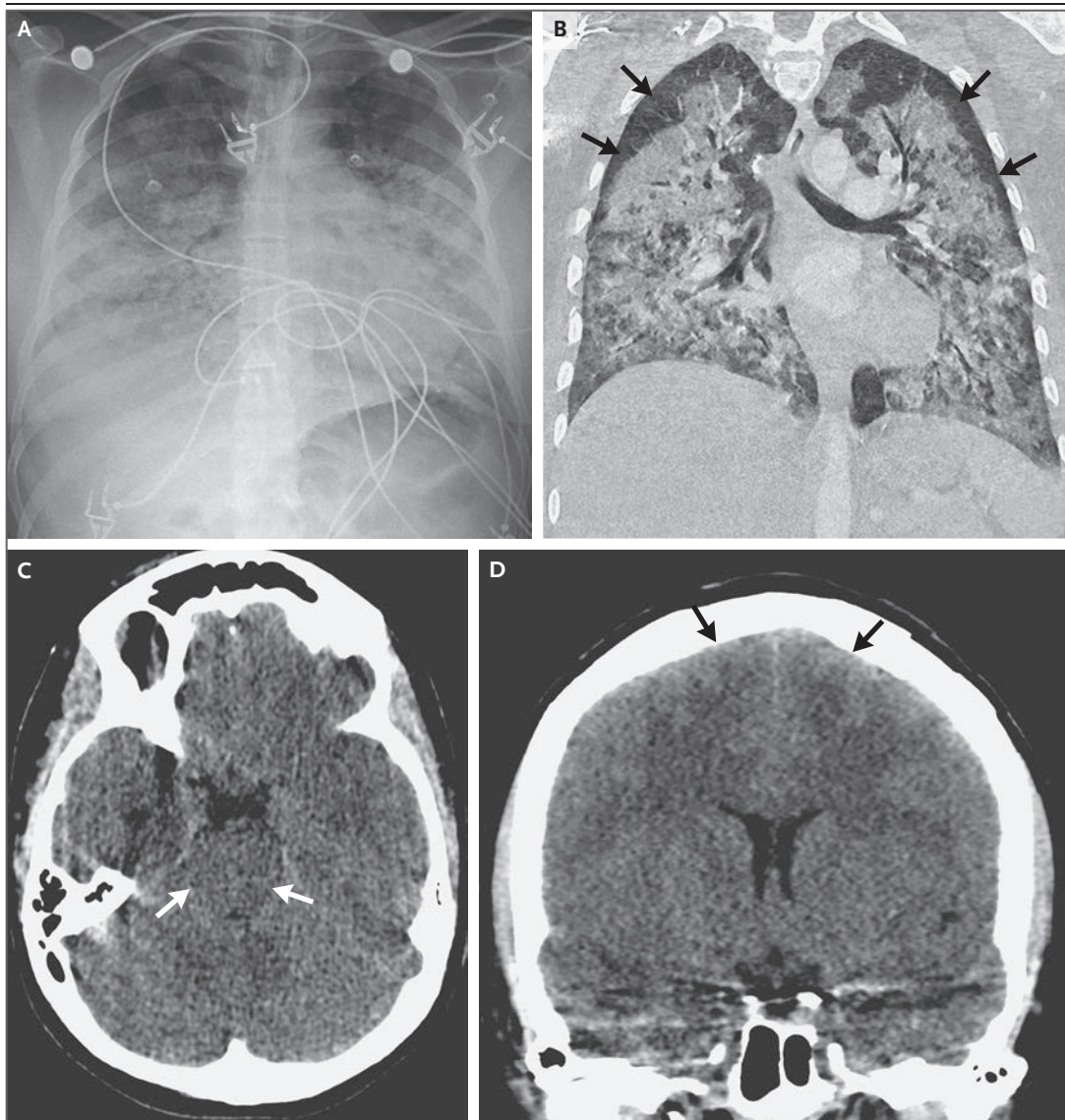


Figure 1. Initial Imaging Studies.

A frontal chest radiograph (Panel A) shows diffuse confluent opacities in both lungs, with relative sparing of the lung apices. A coronal CT image of the chest (Panel B) shows diffuse mixed ground-glass and consolidative opacities involving all lobes of the lungs, with relative sparing of the lung apices and of the peripheral subpleural lung (arrows). An axial CT image of the head (Panel C) shows mild diffuse cerebral edema and effacement of the sulci and basilar cisterns (arrows). A coronal reformatted CT image of the head (Panel D) shows effacement of the cerebrospinal fluid spaces extending to the vertex (arrows), with preserved gray–white matter differentiation.

test for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) was negative. The troponin I level increased to 4.31 ng per milliliter (reference range, <0.029). The thyrotropin, carboxyhemoglobin, and lipase levels were normal, and tests for *Legionella pneumophila* and *Streptococcus pneumoniae* urinary antigens were negative.

Later that afternoon, the patient was transferred to the ICU at this hospital. A review of systems was unavailable because intubation was performed on arrival. Medical history that was obtained from family members included iron-deficiency anemia, anxiety disorder, insomnia, and two normal spontaneous vaginal deliveries.

Regular medications included lorazepam as needed for sleep; postoperative medications included tramadol, gabapentin, and oxycodone as needed for pain. There were no known adverse reactions to medication. The patient had attended college and worked in an office. She lived with her husband and children. She drank alcohol intermittently and did not use tobacco or other substances. There was no family history of cardiopulmonary or neurologic disorders.

The temperature was 36.6°C, the heart rate 88 beats per minute, and the blood pressure 118/78 mm Hg while the patient was receiving norepinephrine at a rate of 8 µg per minute. The height was 168 cm and the body-mass index (the weight in kilograms divided by the square of the height in meters) 27.8. The oxygen saturation was 99% while the patient was receiving oxygen through a mechanical ventilator in volume-control mode (tidal volume, 320 ml; positive end-expiratory pressure, 12 cm of water; respiratory rate, 18 breaths per minute; fraction of inspired oxygen, 1.0). The pupils were 2 mm and symmetrically reactive. The lung sounds were coarse. The abdominal incision was well approximated and without drainage; abdominal drains were in place on both sides. A neurologic examination with interruption of sedation showed spontaneous eye-opening but an inability to visually track and to follow commands. Brain-stem reflexes were intact. The legs withdrew equally and briskly in response to pain. The remainder of the examination was normal.

A chest radiograph showed diffuse opacities in both lungs. Levels of electrolytes, urea nitrogen, creatinine, calcium, alanine aminotransferase, bilirubin, creatine kinase, lipase, and ammonia were normal; other laboratory test results are shown in Table 1. Urine human chorionic gonadotropin was undetectable. Urine toxicologic screening was negative except for the presence of opiates. Specimens of blood and urine were obtained for microbiologic culture. An electrocardiogram including posterior leads showed sinus rhythm, rightward axis deviation, diminutive Q waves, and minor nonspecific ST-T-wave abnormalities. An abdominal radiograph showed a nonspecific bowel gas pattern, subcutaneous emphysema, and surgical drains over the pelvis. On bedside cardiac ultrasonography, the left ventricle could not be visualized clearly, but right

ventricular dilatation did not appear to be present. Intravenous vancomycin, cefepime, and metronidazole were administered.

Dr. Mendoza: CT of the head, performed without the administration of contrast material, revealed hypodensities in the left basal ganglia and right cerebellum that were suggestive of infarcts; the previously described cerebral edema had reduced slightly. A CT angiogram of the head and neck was normal. CT of the chest, performed after the administration of intravenous contrast material, revealed a reduction in ground-glass opacities but an increase in consolidations, particularly in the lower lobes. CT of the abdomen and pelvis, performed after the administration of intravenous contrast material, revealed a small anterior abdominal hematoma and diffuse subcutaneous emphysema.

Dr. Shah: A neurology consultation was obtained; aspirin and atorvastatin were administered. On hospital day 2, the findings on neurologic examination worsened, with only sluggish eye-opening and no motor responses to noxious stimuli. Intravenous mannitol at a dose of 1 g per kilogram of body weight was administered.

Dr. Mendoza: Repeat CT of the head, performed without the administration of contrast material, revealed a hypodensity in the left globus pallidus that had not changed from the previous study, as well as a hypodensity in the right cerebellum that was not definitive and was probably an artifact. Magnetic resonance imaging (MRI) of the head showed infarcts within the globi pallidi and smaller punctate infarcts in the cerebral white matter. There were also innumerable diffuse microhemorrhagic foci affecting the cerebral white matter, including the corpus callosum (Fig. 2).

Dr. Shah: Bronchoscopy revealed bloody mucus in the distal trachea (Fig. 3). Serial bronchoalveolar lavage (BAL) performed in the right upper lobe revealed increasingly erythematous effluent.

Lupus anticoagulant and anticardiolipin antibodies were not detected. Levels of C3 and C4 were normal. Tests for antinuclear antibodies and antineutrophil cytoplasmic antibodies were negative. Tests for legionella, streptococcus, and galactomannan urinary antigens and antitreponemal antibodies were negative, as was a repeat nucleic acid test for SARS-CoV-2.

Dr. Yasmin Aghajan: On hospital day 3, the pa-

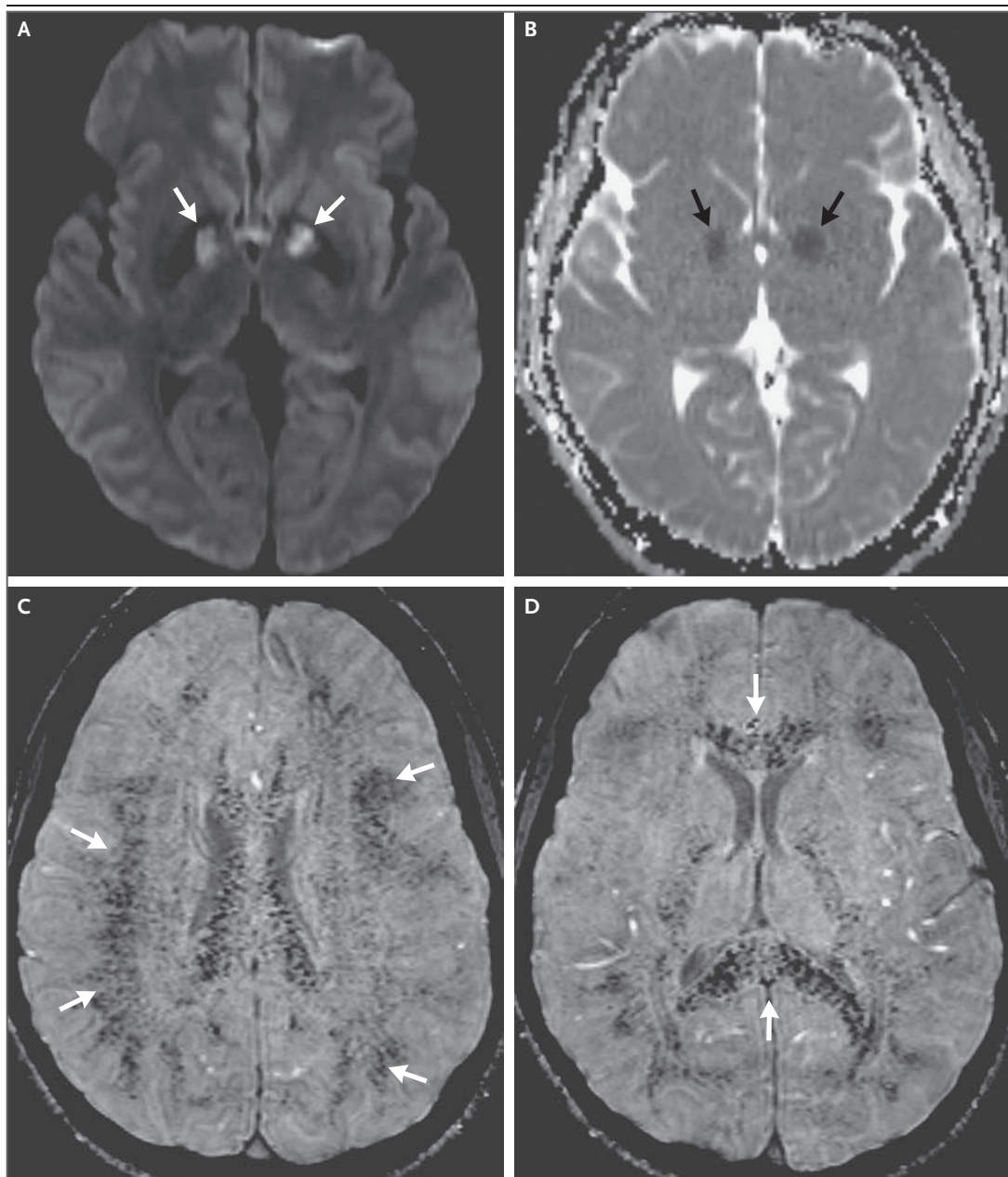


Figure 2. MRI of the Head.

MRI of the head was performed on hospital day 2. An isotropic diffusion-weighted image (Panel A) shows foci of increased signal in the globi pallidi (arrows). A corresponding apparent diffusion coefficient map image (Panel B) shows low values (arrows), a finding that confirms the presence of abnormal restricted diffusion in the globi pallidi. Susceptibility-weighted images (Panels C and D) show diffuse foci of microhemorrhage (signal loss) throughout the cerebral white matter (Panel C, arrows) and the corpus callosum (Panel D, arrows).

tient underwent continuous electroencephalographic monitoring. The study was abnormal because of generalized rhythmic delta activity at a frequency of 1 Hz, which occasionally evolved to generalized periodic discharges superimposed

on a disorganized background with irregular delta and theta slowing. There were no epileptiform abnormalities. Although generalized rhythmic delta activity and generalized periodic discharges are on the ictal–interictal continuum,

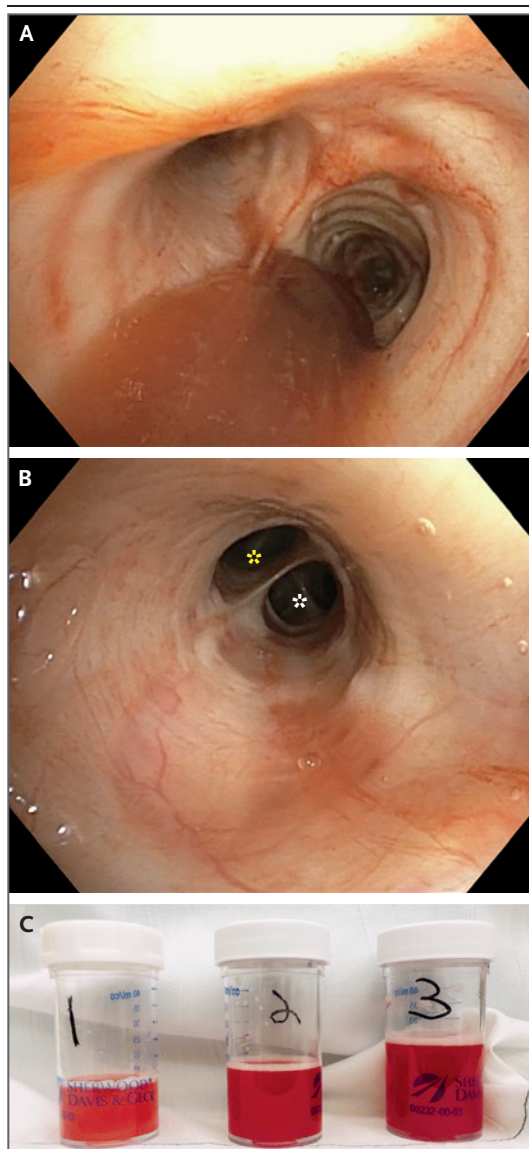


Figure 3. Images Obtained on Bronchoscopy.

Flexible bronchoscopy was performed on hospital day 2. Panel A shows bloody secretions in the airways and along the airway walls, which were easily removed with suctioning at the level of the main carina. Panel B shows the bronchus intermedius with openings to the right middle lobe (yellow asterisk) and the right lower lobe (white asterisk) that have thin bloody secretions, which are representative of the findings throughout the airways. There are no mucosal abnormalities, lesions, or obstructions to the level of the subsegmental airways. Serial bronchoalveolar lavage was performed in the anterior segment of the right upper lobe with the sequential administration of 30 ml of normal saline. Panel C shows the return of increasingly erythematous effluent, a finding consistent with diffuse alveolar hemorrhage.

there were no features suggestive of seizures. The electroencephalographic findings were consistent with encephalopathy and the effects of propofol.

A diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Aran Kadar: This 26-year-old, otherwise healthy woman presented shortly after elective abdominoplasty and liposuction with respiratory failure, diffuse pulmonary opacities, and cerebral edema.

Given the context, two diagnoses come immediately to mind. The first possibility is post-procedural aspiration of gastric contents. Gross aspiration would not necessarily be the cause, because even a small amount of acidic material can lead to widespread lung injury. The second possibility is fat embolism. This rare condition is most often associated with long-bone fracture, but it has been described with other mechanisms of trauma to adipose tissue, including liposuction.¹

Delayed postoperative hypoxemic respiratory failure is often suggestive of pulmonary embolism, but this condition would not be expected to produce such diffuse opacities. This patient's findings on imaging prompt a review of the differential diagnosis for acute alveolar filling processes.

ACUTE ALVEOLAR FILLING PROCESSES

Conditions that cause acute alveolar filling processes can be considered in terms of the substances that rapidly accumulate in the alveoli in the appropriate clinical scenario. The most common substances are water, pus, and blood. Cellular materials such as cancer cells, eosinophils associated with eosinophilic pneumonia, and fibroblasts associated with organizing pneumonia, as well as the surfactant associated with pulmonary alveolar proteinosis, tend to accumulate at a more prolonged or subacute tempo.²

PULMONARY EDEMA

Among the most common causes of acute alveolar filling processes are conditions that lead to pulmonary edema, which can be either cardiogenic or noncardiogenic. Cardiogenic pulmonary edema typically involves acute myocardial dysfunction, which can occur with acute coronary syndrome. Sudden valvular failure and hypertensive emergency can also precipitate acute pulmonary edema. Noncardiogenic pulmonary edema

ma may be considered to be synonymous with acute respiratory distress syndrome, which is most often provoked by infection but can also be seen with exposure to caustic materials, drugs, trauma, or inflammation outside the chest. The widespread distribution of pulmonary opacities in this case favors a noncardiogenic process.

INFECTION

Lower respiratory tract infection can lead to an acute purulent alveolar filling process and can be due to a broad array of pathogens. However, bronchoscopic evaluation of the airways did not reveal purulence, so an infectious process is unlikely.

HEMORRHAGE

Blood that rapidly accumulates in the alveoli typically originates from either airway lesions or diffuse alveolar hemorrhage. Serial BAL can be performed to distinguish between these two origins and may help to further refine the differential diagnosis. In this case, the increase in blood on serial BAL suggests alveolar hemorrhage.

Diffuse alveolar hemorrhage is often associated with vasculitis, especially antineutrophil cytoplasmic antibody–associated disease, although any diffuse alveolar injury can lead to hemorrhage. Effects of drugs, especially anticoagulants, are another frequent cause of alveolar hemorrhage. On the basis of our knowledge of this patient's history, we have no reason to suspect vasculitis, and she was not taking anticoagulants.

To further refine the differential diagnosis, we need to address other relevant aspects of the patient's presentation. Several data points distract from the most likely diagnosis or reflect only the severity of illness without pointing toward the cause. Among these less useful data points is the elevated lactate level, which probably reflects increased sympathetic tone rather than an organ threatened with ischemia. In addition, she had an elevated troponin level, which is a cardiac marker that is elevated in more than 40% of critically ill patients. However, the history, the findings on the electrocardiogram obtained at presentation, and the absence of a wall-motion abnormality on bedside cardiac ultrasonography (albeit with limited visualization) all argue against an acute coronary syndrome.

The patient had an elevated procalcitonin level, a finding that would ideally help to distin-

guish between bacterial infection and other causes. Fat embolism causes an increase in the level of tumor necrosis factor α , which would be expected to increase the procalcitonin level and has been described in case reports of fat embolism.³ This information does not help us to distinguish between the presence of superimposed bacterial infection and aspiration. The elevated B-type natriuretic peptide level may reflect cardiac strain from the severe hypoxemia, but it does not point us toward left ventricular dysfunction as a proximate cause.

A critical additional feature of this patient's presentation is cerebral edema and the later development of a pattern of brain injury with a more embolic appearance. Delayed posthypoxic leukoencephalopathy could produce changes in mental status, but without hypotension or a disruption of cerebral blood flow, it would not typically generate the degree of structural damage observed in this patient.⁴ Carbon monoxide poisoning could produce cerebral edema and focal changes in the globus pallidus,⁵ but the history does not suggest such an exposure.

Because fat droplets are compliant enough to pass through the pulmonary capillary bed and embolize to the central nervous system even in the absence of a patent foramen ovale, the diagnosis of fat embolism syndrome occurring after the liposuction procedure best explains this patient's combination of lung injury, diffuse alveolar hemorrhage, and cerebral injury.

DR. ARAN KADAR'S DIAGNOSIS

Fat embolism syndrome.

PATHOLOGICAL DISCUSSION

Dr. Kartik Viswanathan: The cytopathology laboratory received 35 ml of a blood-tinged BAL specimen. A portion of the BAL specimen was tested with a SurePath liquid-based preparation (Becton Dickinson), and the other portion was applied to a cytospin slide for oil red O (ORO) staining. ORO staining is used to assess for the presence of lipids, which appear red when visualized under light microscopy.⁶

On review of the SurePath slide, there were few benign bronchial cells, abundant acute inflammatory cells, and scattered macrophages with well-

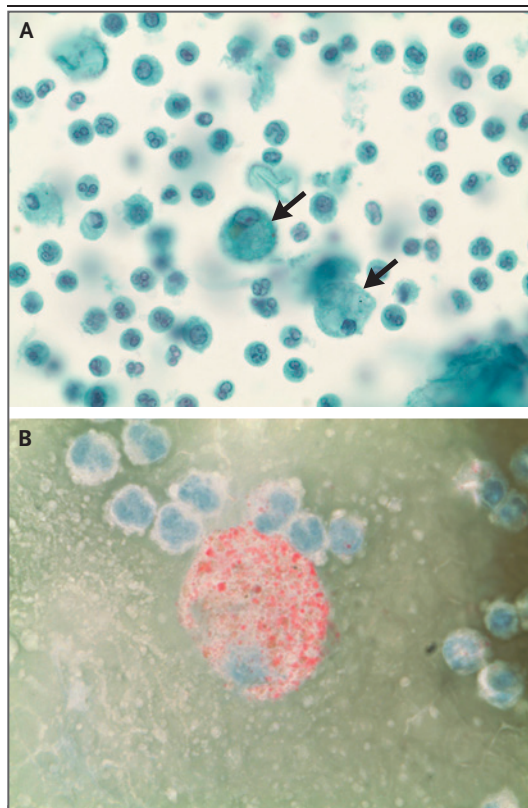


Figure 4. Bronchoalveolar-Lavage Specimens.

A SurePath preparation of a bronchoalveolar-lavage specimen (Panel A) shows scattered alveolar macrophages with well-defined cytoplasmic vacuoles (arrows) amid abundant acute inflammation. An example of oil red O staining (Panel B) shows the red appearance of a lipid-laden macrophage when visualized under light microscopy.

defined cytoplasmic vacuoles (Fig. 4A). On light microscopic examination with ORO staining, the well-defined vacuoles showed red staining, which highlighted the lipid content (Fig. 4B). The ORO staining was scored in 100 macrophages in hot spots, and the scores were summated to generate the lipid-laden macrophage index (LLMI),^{7,8} which was 110. At our institution, ORO staining of BAL specimens is most often used to assess for chronic aspiration in pediatric patients with airway obstruction and in patients who have undergone lung transplantation as a part of their postoperative surveillance. As a point of reference, an LLMI of more than 90 is considered to indicate high risk for aspiration in these patients.^{7,8}

Data regarding the use of ORO staining of BAL specimens as an indicator of fat embolism are limited. A few studies have reported increased

lipid-laden macrophages with ORO staining in BAL specimens from patients with trauma and patients with sickle cell disease who had fat embolism⁹⁻¹¹; however, the correlation has not been consistent in other studies.^{12,13} Moreover, various conditions have been associated with increased ORO-positive lipid-laden macrophages, including infectious pneumonia, aspiration, gastroesophageal reflux disease, the use of glucocorticoid treatment, and human immunodeficiency virus infection.¹⁴ Thus, the presence of increased ORO-positive lipid-laden macrophages is not specific for fat embolism.

Other notable potential challenges in interpreting the results of ORO staining of BAL specimens include variation in scoring between observers, variation in scoring systems (LLMI vs. percentage of ORO-positive alveolar macrophages), staining quality, air-bubble formation, and the presence of hemosiderin and anthracotic pigment in alveolar macrophages.⁸ Overall, the LLMI was increased in this patient's BAL specimen; however, given the relatively low specificity of this test, the LLMI would need to be interpreted in conjunction with clinical and radiologic findings.

PATHOLOGICAL DIAGNOSIS

Cytologic findings consistent with fat embolism.

DISCUSSION OF MANAGEMENT

Dr. Gregory Piazza: Fat embolism syndrome is characterized by a pattern of clinical findings that develops after an insult associated with the release of fat into the circulation.¹⁵ It is most often observed after a traumatic injury, typically a long-bone fracture of the leg. The incidence is estimated to range from 0.54% in patients with an isolated femoral fracture to 1.29% in patients with multiple fractures, including a femoral fracture.¹⁶ The odds that fat embolism syndrome will develop after a delay in primary bone reduction and fixation are twice the odds in the absence of such a delay (odds ratio, 2.21; 95% confidence interval, 1.16 to 4.23).¹⁷

Nonorthopedic causes of fat embolism syndrome are exceedingly rare and include pancreatitis, sickle cell disease, alcoholic liver disease, bone marrow harvest or transplantation, and liposuction. Liposuction is an increasingly common

cosmetic surgery, with 305,856 procedures performed in 2015.¹⁸ During liposuction, rupture of small blood vessels may allow lipid microfragments to reach the venous circulation and in some cases the arterial circulation, resulting in end-organ injury. Coronary occlusion caused by fat embolism after liposuction has been reported.¹⁹

Patients with fat embolism syndrome typically present with multiorgan dysfunction within 12 to 72 hours after the initial insult. The classic triad of manifestations is hypoxemia, neurologic abnormalities, and petechiae.²⁰ Pulmonary symptoms may progress from dyspnea and tachypnea to respiratory failure. Neurologic manifestations include focal deficits, confusion, lethargy, and ultimately coma. Hematologic findings include petechiae, thrombocytopenia, anemia, and disseminated intravascular coagulation. Patients with fulminant cases of fat embolism syndrome may present with right ventricular or biventricular failure, acute respiratory distress syndrome, or shock, or the illness may lead to death.

The diagnosis of fat embolism syndrome is most often made clinically, on the basis of recognition of an appropriate predisposing insult, compatible symptoms and physical findings, and supportive imaging and laboratory studies.¹⁵ Chest radiographs may appear normal or show interlobular septal thickening, patchy ground-glass or nodular opacities, or diffuse bilateral infiltrates consistent with acute respiratory distress syndrome. MRI of the head can reveal diffuse punctate hyperintense lesions (known as the starfield pattern) on diffusion-weighted sequences. In a BAL sample, staining of more than 30% of alveolar cells for neutral fat strongly supports a clinical diagnosis of fat embolism syndrome.

Although there is no standard test for the diagnosis of fat embolism syndrome, diagnostic criteria have been proposed in order to provide support in clinical decision making. The most highly cited criteria are those by Gurd.²¹ Major criteria include petechial rash, respiratory symptoms with abnormalities on imaging, and signs involving the central nervous system that are unrelated to trauma or another condition. Minor criteria include tachycardia, pyrexia, retinal changes (fat or petechiae), renal abnormalities (oliguria, anuria, or lipiduria), acute thrombocytopenia, acute anemia, an increased erythrocyte sedimentation rate, and fat globules in sputum. Typically,

the diagnosis of fat embolism syndrome requires the presence of one major and at least four minor criteria. On the basis of this patient's presentation with severe hypoxemia, diffuse bilateral pulmonary opacities, cerebral edema and encephalopathy, tachycardia, and hypotension after liposuction, she meets the clinical criteria for fat embolism syndrome.

DISCUSSION OF ICU MANAGEMENT

Dr. Peggy S. Lai: The mainstay of treatment for fat embolism syndrome is supportive care. In this case, we administered mechanical ventilation with the use of a lung-protective approach; vasopressors with norepinephrine to maintain a mean arterial blood pressure of more than 65 mm Hg; and empirical broad-spectrum antibacterial therapy with vancomycin, cefepime, and metronidazole, which was subsequently narrowed to a 7-day course of ceftriaxone.

On hospital day 1, the patient underwent intubation and had severe hypoxemia, for which she received 100% supplemental oxygen, with a ratio of partial pressure of arterial oxygen to fraction of inspired oxygen of 115. The hypoxemia rapidly abated; she then received 30% oxygen on hospital day 2 and was transitioned to pressure support ventilation on hospital day 3. The primary obstacle that was preventing extubation was her altered mental status. She also had severe coughing episodes with post-tussive vomiting when doses of sedative medications were lowered and administered less frequently. Extubation was ultimately performed on hospital day 6, and she was transferred out of the ICU on hospital day 7.

DISCUSSION OF NEUROLOGY MANAGEMENT AND PROGNOSIS

Dr. Erica C. Camargo: This patient had ischemic strokes and innumerable small hemorrhagic infarcts, features that were suggestive of an inflammatory hemorrhagic leukoencephalopathy. There are no evidence-based specific therapies for the management of cerebral fat embolism. Antiplatelets and statin therapy were initiated in this patient because of concerns about ischemic stroke but were later discontinued. The mainstay of treatment was systemic supportive care, includ-

ing strict hemodynamic and metabolic control. Patients with cerebral fat embolism undergo intensive neurologic monitoring to assess for neurologic decline and the need for neurologic interventions. Although seizures and status epilepticus may occur, there is no role for prophylactic anti-epileptic medications. There is insufficient evidence to support the use of glucocorticoid therapy in the treatment of cerebral fat embolism, although a meta-analysis reported its potential effectiveness in preventing fat embolism syndrome and hypoxemia in patients with long-bone fractures.²²

This patient's condition improved after her admission to the hospital, and progressive neurologic recovery was sustained after discharge. The neurologic prognosis in patients with cerebral fat embolism varies. Although initial neurologic symptoms — including coma, seizures, status epilepticus, elevated intracranial pressure, and progressive brain injury — can be severe, 72 to 90% of patients may regain functional independence, especially those with a milder burden of disease.^{23,24} In addition, 58% of patients in a coma and those with motor posturing may recover favorably.²⁴ Residual impairments may include minor motor deficits; mild cognitive impairment, especially weaknesses in frontal function and verbal memory; and dysarthria.²³⁻²⁶ Neurologic improvement can occur over the course of several months. It is important to note that cerebral fat embolism does not necessarily imply irreversible structural brain damage, and estab-

lishment of a dire neurologic prognosis early in the course of the disease should be avoided.

Dr. Shah: On hospital day 7, the patient was transferred to the general medicine unit. She had cognitive deficits, with a score on the Montreal Cognitive Assessment of 14, indicating moderate cognitive impairment; scores range from 0 to 30, with higher scores indicating better cognitive function. She was discharged to a rehabilitation facility on hospital day 15. At a 1-month follow-up visit, her score on the Montreal Cognitive Assessment had normalized at 26, and she had regained functional independence.

The Patient: I'm so very thankful for all the nurses and doctors at Mass General. Without their care, I might not have made it. I was inspired by the nurses' kindness. I remember being in the ICU and feeling a bit lonely, and two nurses washed and braided my hair and shaved my legs. It seems so silly, but it's something that I'll never forget. These nurses inspired me to want to be a nurse — I applied to nursing school and got in. Although this illness was a traumatic experience and probably the worst thing that's ever happened to me, it was bearable because of the care I received.

FINAL DIAGNOSIS

Fat embolism syndrome.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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